

Story 1: Pipe Flow Regulation

$$\frac{dy}{dt} + 2y = e^t$$

$$M = e^{\int 2 dt} = e^{2t} \quad (\text{Integrating Factor})$$

$$e^{2t} \left(\frac{dy}{dt} + 2y \right) = e^{2t} \cdot e^t$$

$$\frac{d}{dt} (e^{2t} \cdot y) = e^{3t}$$

$$\int \frac{d}{dt} (e^{2t} \cdot y) dt = \int e^{3t} dt$$

$$e^{2t} \cdot y = \frac{1}{3} e^{3t} + c$$

$$y(t) = \frac{1}{3} e^t + c \cdot e^{-2t}$$

Story 2: Sediment Accumulation in a River

$$\frac{dy}{dx} = xy^2 \quad (\text{Separable Variables})$$

$$\frac{1}{y^2} dy = x dx$$

$$\int \frac{1}{y^2} dy = \int x dx$$

$$-\frac{1}{y} = \frac{x^2}{2} + c' \quad \rightarrow \quad -\frac{1}{y} = \frac{x^2 + 2c'}{2} \quad (2c' = c)$$

$$y(x) = -\frac{2}{x^2 + c}$$

Story 3: Stability of Retaining Walls

$$(x^2+y)dx + (y^2+x)dy = 0 \quad (\text{Exact Differential Equations})$$

$$M(x,y) = x^2+y \quad N(x,y) = y^2+x$$

$$\frac{\partial M}{\partial y} = 1 = \frac{\partial N}{\partial x} = 1 \rightarrow \text{tot. dif. denk.}$$

$$M(x,y) = \frac{\partial u}{\partial x}, \quad N(x,y) = \frac{\partial u}{\partial y}$$

$$u(x,y) = \int M(x,y) dx = \int N(x,y) dy$$

$$\int (x^2+y) dx = \int (y^2+x) dy$$

$$\frac{x^3}{3} + xy + f(y) = \frac{y^3}{3} + xy + f(x)$$

$$u(x,y) = \frac{x^3}{3} + xy + \frac{y^3}{3} + c$$

Story 4: Beam Loading Analysis

$$\frac{dy}{dx} + y = xy^2 \quad (\text{Bernoulli: Differential Equations})$$

$$n=2, \quad P(x)=1, \quad Q(x)=x, \quad v = \frac{1}{y}, \quad \frac{dv}{dx} = -\frac{1}{y^2} \cdot \frac{dy}{dx}$$

$$\frac{1}{y^2} \cdot \frac{dy}{dx} + \frac{1}{y} = x$$

$$-\frac{dv}{dx} + v = x \rightarrow \frac{dv}{dx} - v = -x \quad \mu = e^{\int -1 dx} = e^{-x}$$

$$e^{-x} \left(\frac{dv}{dx} - v \right) = e^{-x} \cdot (-x)$$

$$\frac{d}{dx} (e^{-x} \cdot v) = -e^{-x} \cdot x$$

$$u=x \rightarrow du=dx \\ dv=e^{-x} dx \rightarrow v=-e^{-x}$$

$$\int \frac{d}{dx} (e^{-x} \cdot v) dx = - \int e^{-x} \cdot x dx$$

$$e^{-x} \cdot v = x \cdot (-e^{-x}) - \int -e^{-x} dx = -x \cdot e^{-x} - e^{-x} + c' = -e^{-x}(x+1) + c'$$

$$e^{-x} \cdot v = -e^{-x}(x+1) + c'$$

$$\frac{1}{y} = v = -(x+1) + c' \quad (c' = -c)$$

$$y(x) = -\frac{1}{x+1+c}$$

Story 5: Concrete Pouring Rates

$$\frac{dy}{dt} = 2t, \quad t=0, \quad y=3 \quad (\text{Separable Variables})$$

$$\int dy = \int 2t dt$$

$$y = t^2 + c$$

$$y(t=0) = 3 \rightarrow 0 + c = c = 3$$

$$y(t) = t^2 + 3$$

Story 6: Heat Dissipation in a Processor

$$\frac{dT}{dt} = -k(T - T_a), \quad k > 0 \text{ constant}, \quad T(0) = T_0 \quad (\text{Separable Variables})$$

$$\frac{1}{T - T_a} dT = -k dt$$

$$\int \frac{1}{T - T_a} dT = \int -k dt$$

$$\ln|T - T_a| = -kt + c'$$

$$T - T_a = e^{-kt + c'} \quad (e^{c'} = C)$$

$$T - T_a = e^{-kt} \cdot C$$

$$T(t=0) = T_0 = T_a + e^{-k \cdot 0} \cdot C = T_a + C \rightarrow C = T_0 - T_a$$

$$T(t) = T_a + (e^{-kt})(T_0 - T_a)$$

Story 7: Network Traffic Modeling

$$\frac{dP}{dt} + P = e^{-2t}, \quad P(0) = P_0$$

$$\mu = e^{\int 1 dt} = e^t \quad (\text{Integrating Factor})$$

$$e^t \left(\frac{dP}{dt} + P \right) = e^t \cdot e^{-2t}$$

$$\frac{d}{dt} (e^t \cdot P) = e^{-t}$$

$$\int \frac{d}{dt} (e^t \cdot P) dt = \int e^{-t} dt$$

$$e^t \cdot P = -e^{-t} + C$$

$$P = -e^{-2t} + C \cdot e^{-t}$$

$$P(t=0) = -e^{-2 \cdot 0} + C \cdot e^{-0} = C - 1 = P_0 \rightarrow C = P_0 + 1$$

$$P(t) = -e^{-2t} + (P_0 + 1)e^{-t}$$

Story 8: Gradient Descent for Optimization

$$f(w) = w^2 + 2w + 5, \quad w_{n+1} = w_n - \alpha \frac{df}{dw}, \quad w_0 = 3, \quad \alpha = 0,1$$

$$\frac{df}{dw} = \frac{d}{dw} (w^2 + 2w + 5) = 2w + 2$$

$$w_{n+1} = w_n - \alpha (2w_n + 2)$$

$$w_{n+1} = w_n - 0,1 (2w_n + 2)$$

$$w_{n+1} = 0,8w_n - 0,2$$

$$n=0 \rightarrow w_1 = 0,8 \cdot 3 - 0,2 = 2,2$$

$$n=1 \rightarrow w_2 = 0,8 \cdot 2,2 - 0,2 = 1,56$$

$$n=2 \rightarrow w_3 = 0,8 \cdot 1,56 - 0,2 = 1,048$$

$$w_1 = 2,2, \quad w_2 = 1,56, \quad w_3 = 1,048$$